

Hunting the Quarter-Turn: An Empirical Test of Posited Dihedral Structure in a Language Model’s Representation of Hohfeldian Positions

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Abstract

Hohfeld’s four normative positions (Obligation, Claim, Liberty, No-claim) admit two demonstrated operations: the correlative swap between parties and deontic negation. A machine-checked result establishes that these two operations are commuting involutions and generate only the Klein four-group V_4 ; the order-eight dihedral group D_4 , which adds a quarter-turn cycling the four positions, is posited but has never been demonstrated as a normative operation. We test whether the quarter-turn exists in a language model’s learned representation of normative statements. The key observation is that the quarter-turn has a direct pair corpus: the four-cycle over jural statements about the same scenario. We fit linear representation maps for the swap, the negation, and the cycle with the same ridge estimator, on one surface-template family, and grade the dihedral relations on held-out scenarios expressed in a second template family, so the maps must encode jural structure rather than wording. The verdict rule is stated before the run: the quarter-turn counts as found only if the fitted cycle map squares to the measured negation, satisfies the dihedral relations, and reproduces the non-abelian predictions at errors comparable to the measured V_4 sector’s own self-consistency, with a wrong-target control. In a 0.5B-parameter model the quarter-turn is not found: the fitted cycle map’s square does match the measured negation on the component where negation is linearly represented at all, but the non-abelian signature is absent, and even the measured swap generalizes weakly across surface templates, which bounds the instrument’s power. The V_4 -only conclusion survives its first test in a learned representation, and the experiment yields a side finding: deontic negation is by far the most linearly represented Hohfeldian operation in this model.

1 Introduction

Hohfeld separated four fundamental normative positions: Obligation, Claim, Liberty, and No-claim [1]. Two operations on these positions are demonstrated in practice. The correlative swap s exchanges the parties’ perspectives ($O \leftrightarrow C$, $L \leftrightarrow N$). Deontic negation r^2 negates the normative status ($O \leftrightarrow L$, $C \leftrightarrow N$). A recent machine-checked result, which we call the keystone correction, establishes in Lean 4 with Mathlib that s and r^2 are commuting involutions, that the group they generate is isomorphic to the Klein four-group V_4 , and that the quarter-turn r ($O \rightarrow C \rightarrow L \rightarrow N \rightarrow O$) lies outside the generated subgroup [2]. The dihedral group D_4 of order eight is therefore posited ambient structure: it is licensed only if the quarter-turn is independently demonstrated as a normative operation, and it has not been.

This paper asks a narrower question than “is moral reasoning D_4 -structured”: does the quarter-turn exist as a coherent linear operation in a language model’s learned representation of normative statements? The question is empirical and falsifiable, and both answers are informative. If a fitted cycle map satisfies the dihedral relations out of sample, the posited D_4 acquires its first empirical footing in any substrate, and the place to look for its meaning is Hohfeld’s second square: the power operations (waiver, grant, authorization) that transform first-square positions and are directed rather than involutive. If no such map exists, the keystone’s V_4 -only conclusion extends from the axiomatic level to at least one learned representation.

Two methodological points make the test sharp. First, the quarter-turn has a direct pair corpus. Prior discussion treated r as an abstract missing generator; we observe that the four-cycle over jural statements about the same scenario is itself a supervised mapping task, so $\hat{\rho}(r)$ can be fitted exactly as $\hat{\rho}(s)$ and $\hat{\rho}(r^2)$ are. Second, surface form is controlled: maps are fitted on one template family and graded on a second, over held-out scenarios, so a map that merely memorizes template wording fails.

2 Background

2.1 The keystone correction

Earlier materials in this program claimed the full dihedral D_4 as the symmetry group of the Hohfeldian square. The July 2026 keystone correction retracted that: only s and r^2 are demonstrated operations, they commute, and the group they generate is V_4 . The correction is machine-checked (`formal/HohfeldV4.lean`: commuting involutions, four-element closure isomorphic to V_4 , quarter-turn exclusion, and well-definedness of the ambient D_4 machinery) and is recorded, with the measured-versus-posed discipline it imposes on every downstream document, in the program’s concept registry (`docs/CONCEPT_REGISTRY.md`). This paper is the registry’s discipline applied forward: the posited operation is tested rather than assumed.

2.2 Linear structure in learned representations

Linear maps between representations of systematically related inputs are a standard instrument: relational offsets [3], linear probes [4], and Procrustes or ridge alignment between representation spaces [5]. Our estimator is the regularized ridge map used by the I-EIP monitoring program for exactly this purpose [6]. The contribution here is not the estimator but the group-theoretic test battery applied to its outputs.

3 Method

3.1 Corpus

1,600 scenarios are generated from templated (agent, patient, action) triples. Each scenario is expressed in all four Hohfeldian positions, in two surface-template families that differ in wording but not jural content (eight statements per scenario). Family A: “{A} must {act} for {B}” (O); “{B} is entitled to have {A} {act}” (C); “{A} is free not to {act} for {B}” (L); “{B} has no claim that {A} {act}” (N). Family B rephrases each position (duty to, holds a claim against, under no duty, cannot demand). Templated English is a stated limitation (Section 6).

Test (0 = perfect, 1 = no better than identity)	Layer 12	Layer 18
fit $\hat{\rho}(s)$ (swap, measured op)	0.883	0.931
fit $\hat{\rho}(r^2)$ (negation, measured op)	0.484	0.532
fit $\hat{\rho}(r)$ (cycle, posited op)	0.735	0.830
$\hat{\rho}(s)\hat{\rho}(r^2) \rightarrow sr^2$	0.525	0.583
$\hat{\rho}(r^2)\hat{\rho}(s) \rightarrow sr^2$	0.533	0.589
$\hat{\rho}(r)^2 = \hat{\rho}(r^2)$	0.488	0.481
$\hat{\rho}(s)\hat{\rho}(r)\hat{\rho}(s) = \hat{\rho}(r)^3$	0.682	0.768
$\hat{\rho}(r)\hat{\rho}(s) \rightarrow rs$ (non-abelian target)	1.036	1.000
$\hat{\rho}(s)\hat{\rho}(r) \rightarrow sr$	0.891	0.985
$\hat{\rho}(r)\hat{\rho}(s) \rightarrow sr$ (wrong-target control)	0.903	0.940

Table 1: Held-out normalized errors, evaluation scenarios in the held-out template family. Identity-target rows ($\hat{\rho}(s)^2 = I$, $\hat{\rho}(r^2)^2 = I$, $\hat{\rho}(r)^4 = I$) are omitted: the normalization denominator is zero when the target equals the source, an instrument limitation of this run’s error definition.

3.2 Representations and maps

Last-token hidden states of an open-weights causal language model (Qwen2.5-0.5B, layers 12 and 18 of 24; layer 24 reported as an anchor) represent each statement. For each candidate operation g with state mapping m_g , a linear map $\hat{\rho}(g)$ is fitted by ridge regression from $h(x_{st})$ to $h(x_{m_g(st)})$ over training scenarios in template family A, pooled over all four source positions. Three maps are fitted identically: $\hat{\rho}(s)$ from the swap pairs, $\hat{\rho}(r^2)$ from the negation pairs, and $\hat{\rho}(r)$ from the cycle pairs $O \rightarrow C \rightarrow L \rightarrow N \rightarrow O$. The training split exceeds the representation dimension, and held-out fit quality is reported.

3.3 Relation tests and the verdict rule

All tests are graded on held-out scenarios expressed in template family B. For a composition $M_k \cdots M_1$ with predicted state mapping m , the error is

$$E = \frac{\sum_{st} \|M_k \cdots M_1 h(x_{st}) - h(x_{m(st)})\|^2}{\sum_{st} \|h(x_{st}) - h(x_{m(st)})\|^2},$$

so $E = 1$ means the composition predicts the target no better than leaving the source representation unchanged, and $E = 0$ is exact. The V_4 sector supplies the yardstick: $E[\hat{\rho}(s)^2 = I]$, $E[\hat{\rho}(r^2)^2 = I]$, and the commutation tests. The rehabilitation tests are $E[\hat{\rho}(r)^2 = \hat{\rho}(r^2)]$ (the fitted cycle squares to the measured negation), $E[\hat{\rho}(r)^4 = I]$, and the conjugation $E[\hat{\rho}(s)\hat{\rho}(r)\hat{\rho}(s) = \hat{\rho}(r)^3]$. The non-abelian signature is graded by target: from the same source position, $\hat{\rho}(r)\hat{\rho}(s)$ and $\hat{\rho}(s)\hat{\rho}(r)$ must land on different D_4 -predicted targets (for example $rs(O) = L$ while $sr(O) = O$), and a wrong-target control checks that $\hat{\rho}(r)\hat{\rho}(s)$ does not fit the sr targets equally well.

The verdict rule, stated before the run: the quarter-turn is found only if the rehabilitation tests land at errors comparable to the V_4 yardstick rows and the non-abelian signature separates from its wrong-target control, out of sample in both scenario and template.

4 Results

Table 1 reports the held-out normalized errors (committed record: `hunt_result.json`; 2,312 statements per position per family after generation, 1,120 training scenarios, 480 evaluation scenarios).

Verdict, per the pre-stated rule: the quarter-turn is not found. The decisive row is the non-abelian signature. $\hat{\rho}(r)\hat{\rho}(s)$ predicts its D_4 target rs no better than leaving the representation unchanged (1.036 and 1.000), and it fits the wrong target sr slightly better (0.903 and 0.940) than the right one. Whatever the fitted cycle map is doing, it does not carry the order-dependence that separates D_4 from V_4 .

Two further findings frame the verdict. First, the yardstick itself is weak: the measured correlative swap barely generalizes across surface templates (0.883 and 0.931), while deontic negation is by far the most linearly represented Hohfeldian operation in this model (0.484 and 0.532). The instrument therefore bounds what it can certify: it cannot find D_4 , and it also cannot strongly certify V_4 as a linear structure at these layers; the negation axis is the one operation this representation supports well. Second, one fragment of dihedral structure does appear: $\hat{\rho}(r)^2$ matches the negation targets exactly as well as the directly fitted $\hat{\rho}(r^2)$ does (0.488 versus 0.484 at layer 12; 0.481 versus 0.532 at layer 18). On the component of the representation where negation is linear at all, the fitted cycle behaves as its square root. It simply carries no non-abelian structure with it.

5 What a found quarter-turn would and would not mean

A linear operator in one model’s representation space is evidence about that model’s learned geometry, not about human normative practice. A found quarter-turn licenses a next question, not a conclusion: whether power operations (waiver, grant, authorization), which transform Hohfeldian positions and are directed rather than involutive, realize the same map on natural text. A missing quarter-turn is evidence that the V_4 -only structure is not an artifact of the axiomatic presentation, since a model trained on natural language also fails to support the larger group.

6 Scope and limitations

Templated English, not court prose; one model at one scale; linear maps only (a nonlinear quarter-turn would be missed); last-token pooling; the cycle corpus teaches the mapping task, so a found $\hat{\rho}(r)$ demonstrates representability rather than spontaneous use. The consumer-relative extension (whether the relations hold in a downstream readout’s metric where they fail in the raw metric, or conversely) is a designed follow-up, not part of this run.

7 Conclusion

The posited quarter-turn does not appear as a coherent linear operation in this model’s representation of Hohfeldian statements: the non-abelian signature is absent, and the pre-stated verdict rule returns not found. The keystone’s V_4 -only conclusion stands, now with one empirical test behind it; the constructive residue is that negation is the operation learned representations support best, and that any future hunt should use a stronger instrument (per-input nonlinear maps, natural rather than templated jural text, and Hohfeld’s power operations as the candidate corpus).

References

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